Ecological assets behind improving accounts: a boundary-matched test of biomass-change identifiability

Amin Abaee
Independent Researcher, Tehran, Iran
amin_abaee@ut.ac.ir
ORCID: 0000-0002-0019-1842

Abstract

Biocapacity accounts and physical ecological assets describe distinct quantities. Testing composition masking therefore requires matched observation domains and an identifiable physical comparison. We assess national account availability and test what matched biomass and mapped-cover observations identify within a fixed Brazilian footprint. Of 236 country and territory identifiers, 183 satisfy the 2015–2023 account criteria; all eight countries selected for physical screening pass those criteria, but none is admitted to a national physical-change comparison. On the 61,830 ha local footprint, the auxiliary whole-cell biomass point-map total increases by 65.24×10^3 Mg, while positive and negative local differences sum to 142.21×10^3 and 76.97×10^3 Mg: 70.23% of absolute local differences cancels in the aggregate. Constructive allocations preserve every observed native-cell total and mapped fraction yet permit either positive or negative group-specific change. The coherent joint set for the nested groups intersects all four strict sign quadrants. Comparing nine nested mass-observation scales on identical support, native-cell information narrows the unrestricted difference-set width by 49.05% for mapped formations and 55.93% for the separately defined formations-plus-savanna domain, but identifies neither sign. A density-contrast sensitivity family loses strict positive-sign identification at assumed within-cell ratio bounds of 1.4555 and 1.9495, respectively; neither bound is empirically established. These data-anchored identification and aggregation results condition on the point maps, not calibrated ecological change. Inspected reference artifacts do not provide compatible paired local endpoints. The results specify why increasing aggregates are insufficient and what additional allocation information could help; they do not establish national account–asset divergence, conversion associations or the occurrence of ecological composition masking.

Keywords: Ecological Footprint; Biocapacity; Natural capital accounting; Above-ground biomass; Partial identification; Subpixel biomass allocation; Land-use change.

1 Introduction

Biocapacity is expressed in global hectares, combining physical area with productivity-based factors. Above-ground biomass is a mass of specified living woody material within a geographic support. These quantities are related through ecological production but are not interchangeable measures of ecosystem assets. The distinction between natural-capital stocks and accounting flows has an established basis in the Ecological Footprint literature (Mancini et al., 2017). A quantitative assessment requires separate observation operators and an explicit treatment of their domains.

Composition-dependent land-use models show that transfers towards a land class with a higher annual contribution per hectare can increase a model account while reducing another land class's area (Abaee, 2026). Such a mechanism motivates an empirical comparison but does not determine the sign of an observed forest-stock trend. Existing European forest-condition

accounting also demonstrates the importance of spatially heterogeneous changes within an aggregate (Maes et al., 2023). Forest condition, extent, biomass and regeneration remain distinct ecological attributes.

The ecological hypothesis motivates a sequence of measurement and inference requirements. The present study tests whether the available information identifies the required physical sign, through four questions:

1. Which national account comparisons satisfy availability criteria, and which physical-support or reference requirements remain unmet?

2. Within a matched local footprint, can an increasing whole-cell biomass point-map total rule out a decline in a separately defined mapped-cover group?

3. How do spatial aggregation and restoration of native-cell mass information change observable offsets and the width or sign of group-change identification sets on identical support?

4. Within an explicit density-contrast sensitivity family, what additional allocation restriction would identify a positive group-change sign?

The account analysis retains all directions of account change, and the local footprint is selected by geometry rather than outcomes. The broader comparisons of independently validated forest stocks, account–asset divergence, conversion associations and reporting accuracy remain subsequent ecological questions, with their admission requirements specified in the Supplementary Information. They are not inferred from conditional point-map arithmetic.

2 Materials and methods

2.1 Sources and observational domains

The account source is the National Ecological Footprint and Biocapacity Accounts, Edition 2026, results and metadata version 1.0 (Lo et al., 2026). The primary comparison interval is 2015–2023. The secondary comparison domain is 2010–2020, conditional on compatible endpoints and separately justified cross-sensor change uncertainty. Absolute total biocapacity is the principal account measure. Terrestrial, excluding-built-up and forest-component accounts, together with separately labelled per-person values, define alternative account domains. The account's own area fields belong to its input lineage and are not independent physical validation measurements.

The biomass source is ESA Climate Change Initiative Biomass version 7.0 (Santoro and Cartus, 2026). The product provides above-ground woody biomass density estimates for 2005–2012 and 2015–2024. Metadata and coordinate inspection used the nominal 50 km aggregate and the 2010–2020 change product. Product definitions and uncertainty interpretation follow the Product User Guide, Algorithm Theoretical Basis Document, end-to-end uncertainty budget and validation report (CCI Biomass, 2026, 2025a,b, 2026b). Bounded native-grid observations at 2015 and 2023 distinguish missing, observed-zero and valid values; they do not estimate national coverage or biomass change. The 2010–2020 example describes product structure; it is not a change-validation dataset for the primary interval.

The observation system combines C- and L-band radar retrievals with structural calibration and land-cover-dependent processing. The C-band branch is generated at 150 m spacing and resampled to the 100 m L-band grid. Interannual refinement of merging weights, including land-cover-specific exceptions, improves consistency but does not establish independent endpoint errors or independent 100 m observations. Land-cover information affects calibration, masking of selected classes and merging decisions; the product does not impose a universal forest mask over vegetated land. Published zero estimates are retained separately from missing observations without interpreting them as independent evidence of absent woody vegetation (CCI Biomass, 2025a, pp. 57–59, 82–83, 87, 91, 95, 101–102).

Eight countries were selected for evaluation of geographic and measurement compatibility: Brazil, Indonesia, Democratic Republic of the Congo, Cameroon, Canada, Finland, France and Australia. This purposive selection provides environmental and logistical contrasts rather than a probability sample. Country-level account availability and physical observation compatibility are assessed separately. The full set of account identifiers is retained in the availability analysis, irrespective of any expected direction of ecological change.

2.2 Territorial correspondence and observation support

Country screening uses the full published geoBoundaries gbOpen ADM0 geometries, identified by fixed repository commits and source hashes (Runfola et al., 2020). Their represented years range from 2016 to 2021; a common geometry across endpoints defines the screening support, not evidence that legal boundaries or account conventions were unchanged. Quantized source geometries with invalid polygon topology are repaired computationally, with the original objects and repair indicators retained. These operations do not certify positional or territorial accuracy.

Polygon areas and their intersections with the native biomass domain, 80°N to 60°S, are calculated on WGS84. Geographic linework is densified to at most 0.02° before geodesic area calculation to approximate clipped parallels. These polygon areas may contain inland or jurisdictional waters and are not forest-area measurements. Forest support, observed zeros, missing observations and converted land require distinct definitions. ISO identifiers and national account availability do not, by themselves, establish common territorial support for all components, particularly fishing accounts. The Supplementary Information distinguishes geographic domain coverage from valid observations and analytical admission.

2.3 Account availability and numerical consistency

Account availability requires a record for every year in 2015–2023, publication flags that permit the relevant biocapacity quantities, and finite, nonnegative required values. Population must be positive. Required fields comprise the five biocapacity components, their total, forest and cropland areas, and population. Flags `+++`, `++` and `+` satisfy the publication criterion; their different trade and consumption detail does not imply a ranking of physical measurement precision. Flags indicating unpublished, missing or estimated-only information do not satisfy the criterion.

The sum of the five published components is compared with the reported total. A tolerance of 3 gha accounts for rounding of five component values and their total. This is a numerical-consistency tolerance, not an ecological threshold. Duplicate country–year keys, missing years and missing required fields are evaluated explicitly. The Supplementary Information specifies the rules and country results.

2.4 External land-cover support

The Brazilian support comparison uses MapBiomas Collection 11 annual land-cover and land-use maps at nominal 30 m spacing for 2015 and 2023 (MapBiomas Project, 2026). The footprint is the same geometry-selected 256 × 256 native biomass-cell rectangle used for observation-support assessment; it is not selected using forest cover or biomass changes and is not a national probability sample. Both cover endpoints use the same published collection and geographic grid.

The operational formations group comprises Forest Formation (3), Mangrove (5), Floodable Forest (6) and Wooded Sandbank Vegetation (49). Savanna Formation (4), Flooded Savanna (7) and Forest Plantation (9) remain separately identified. A formations-plus-savanna quantity is reported as a different cover grouping, not an uncertainty interval around the formations target. These mapped classes do not automatically identify all forest land use, temporarily unstocked forest or ecological assets. The official class definitions are biome-specific.

For each native biomass cell, group fractions are calculated from the WGS84 areas of intersections with finer mapped pixels. Geographic spacing is not treated as equal-area sampling, and

missing or unclassified support is retained separately rather than renormalized away. Published classifications may incorporate temporal gap filling and filtering, so complete label coverage does not imply a direct, contemporaneous observation at every pixel. The Supplementary Information specifies the area-overlap calculation and observation limits.

2.5 Physical support and stock decomposition

For a defined support Ω and a density referring to that same support, stock is

$$Q_t = \int_\Omega d_t(x)\,\mathrm{d}A(x),$$

where d_t is a biomass density with a specified area denominator. Geographic raster cells require latitude-dependent area weights. The computational method integrates meridian–parallel rectangles on the WGS84 ellipsoid; dateline crossings are divided into separate longitude intervals.

For positive forest areas A_0, A_1 and compatible stocks Q_0, Q_1, let $D_t = Q_t/A_t$. The symmetric decomposition

$$\Delta Q_{\text{extent}} = (A_1 - A_0)(D_1 + D_0)/2, \tag{1}$$
$$\Delta Q_{\text{density}} = (D_1 - D_0)(A_1 + A_0)/2 \tag{2}$$

sums exactly to $Q_1 - Q_0$. The contributions describe changes in aggregate area and mean density; they do not identify causal effects. Zero-area endpoints require a separately defined treatment because forest-only density is undefined there. No physical stock decomposition is estimated from the account area fields alone.

2.6 Conditional within-cell biomass allocation

Native CCI AGB is a full-pixel density, so a finer mapped group fraction does not allocate biomass within a mixed cell. A conditional allocation calculation holds the published point maps, grid and epochs fixed and allows any nonnegative within-cell biomass distribution, without a finite subcell density cap or temporal allocation constraint. These assumptions define a minimal algebraic model, not a complete ecological or measurement-error model.

Let $q_{it} = A_i \widehat{d}_{it}$ be the auxiliary full-cell biomass mass, P_t^g the cells entirely mapped as group g, and H_t^g all cells with a positive-area intersection with that group. The compatible aggregate allocation set is

$$\widehat{\mathcal{I}}_t^g = [L_t^g, U_t^g], \qquad L_t^g = \sum_{i \in P_t^g} q_{it}, \quad U_t^g = \sum_{i \in H_t^g} q_{it}.$$

Integer counts of intersecting finer pixels determine these categories; no near-zero/near-one ecological fraction threshold is imposed. All cells remain in the target footprint. Using pure cells to anchor the lower bound is not a persistent-pure-only analysis sample. Endpoint sets imply the allocation-difference set $[L_{2023}^g - U_{2015}^g,\, U_{2023}^g - L_{2015}^g]$ under the stated absence of temporal restrictions. The Supplementary Information establishes attainability within this minimal model. The sets are not confidence intervals or guaranteed bounds on true forest biomass: map error, density error, epoch mismatch and additional ecological constraints are not represented.

2.7 Spatially offsetting local changes

For local stock changes ΔQ_i on a common observed support, define

$$G = \sum_i [\Delta Q_i]_+, \qquad L = \sum_i [-\Delta Q_i]_+,$$

with net change $G - L$, offset mass $\min(G, L)$ and cancellation fraction

$$K = 1 - \frac{|G - L|}{G + L}, \qquad G + L > 0. \tag{3}$$

K is undefined when $G + L = 0$. The components are positive and negative local net changes, not gross biological growth, recruitment, mortality or harvest. Their values depend on spatial support and measurement error. Interpretation therefore requires effective-resolution and zero-change error-model assessment. Temporal comparisons require repeated observations at identifiable dates rather than interpolated or model-generated annual independence.

2.8 Matched-support tests of information sufficiency

The Brazilian analysis compares information sets on exactly the same footprint, endpoints, area weights and mapped domains. The native-cell masses q_{it} are first summed over nested, non-overlapping square blocks with sides $s = 1, 2, 4, \ldots, 256$ native cells. All 65,536 cells remain included at every scale. This operation coarsens the available mass observations; it neither changes the cover classes nor creates observations at a different effective sensor resolution. The calculations are exploratory analyses of a fixed, geometry-selected footprint, not a probability sample or a preregistered ecological test.

For a block b, let $q_{bt} = \sum_{i \in b} q_{it}$. A block is group-pure only if every constituent cell is certified pure by integer overlap counts, and group-present if any constituent cell has positive-area group overlap. Conditional on observing only the block mass, the sharp endpoint set is

$$\widehat{\mathcal{I}}^{g}_{t,s} = \left[\sum_{b \in P^{g}_{t,s}} q_{bt}, \sum_{b \in H^{g}_{t,s}} q_{bt} \right]. \tag{4}$$

The difference set uses cross-endpoint extremes as above. Finer mass observations impose additional constraints, so the sets are nested under coarsening. Width reduction relative to the one-window observation quantifies an increment in identification information, not classification accuracy, confidence coverage or ecological predictive performance. Spatial cancellation is calculated separately from the same block-mass differences using Eq. 3; it describes offsets in published point maps, not independently measured biological gains and losses.

To establish attainability rather than infer meaning from wide intervals alone, two extremal allocations are constructed for each mapped group. The minimum-change witness allocates all mixed-cell mass to the group in 2015 and none in 2023; the maximum-change witness reverses those choices. Pure cells retain all their mass and absent cells none. Each witness preserves every observed native-cell total and every mapped fraction, with nonnegative complement mass. They are separate hypothetical allocations under the stated model, not estimated trajectories or jointly imposed extrema for overlapping groups. The Supplementary Information gives the construction and nesting argument.

Coherent nested-group allocation. Formations (A), added savanna (S) and their complement (O) form a disjoint partition; formations plus savanna is $H = A + S$. A joint allocation uses one nonnegative three-component mass partition in every cell and endpoint. Its sharp difference set in coordinates $(a, h) = (\Delta Q^{A}, \Delta Q^{H})$ is

$$\mathcal{D}_{\text{joint}} = \{(a, h) : a \in \widehat{\mathcal{D}}^{A},\ h \in \widehat{\mathcal{D}}^{H},\ h - a \in \widehat{\mathcal{D}}^{S}\}. \tag{5}$$

The added-savanna interval is calculated from its own integer overlap counts. This joint set imposes coherence omitted by the Cartesian product of separate group intervals. The Supplementary Information establishes sharpness and describes four sign-quadrant witnesses reconstructed to preserve every native-cell mass and available component.

2.9 Sensitivity to within-cell density contrast

A separate sensitivity family asks how strong an additional allocation restriction must be to identify a sign. In a mixed cell with group fraction f, let the nonnegative component densities be d_g and d_h. Impose $d_g \le R d_h$ and $d_h \le R d_g$ for a common assumed $R \ge 1$ across mixed cells and both endpoints. With full-cell mass q, the sharp group-mass bounds are

$$q\frac{f}{f + R(1-f)} \;\le\; s_g \;\le\; q\frac{Rf}{Rf + 1 - f}. \tag{6}$$

Pure and absent cells retain their topologically certified endpoint allocations. The $R = 1$ limit is homogeneous allocation, whereas $R = \infty$ recovers the unrestricted primary model. No finite R is inferred from the data or adopted as an ecological fact. The zero of the aggregate difference lower bound locates the loss of strict positive-sign identification within this sensitivity family. A finite bound would require independent justification before supporting a physical conclusion; this sensitivity calculation does not calibrate measurement error or supply the missing paired reference information.

2.10 Uncertainty, validation and information comparisons

The relevant error structure includes temporal and spatial covariance, systematic bias, land-transition classification error and incomplete coverage. An assessment of biomass levels does not establish the accuracy of biomass change. Independent forest-to-cropland observations require a justified reference design and transition-specific uncertainty (Olofsson et al., 2014).

CCI's uncertainty model already propagates correlated retrieval errors within annual image stacks and spatially correlated errors in aggregated products. Those terms must be distinguished from cross-year covariance of the final maps. In particular, the L-band within-stack correlation assumption of 0.5 is not an estimate of correlation between the 2015 and 2023 endpoint errors. The uncertainty budget's change-variance expression sums endpoint variances without a cross-year term; this omission does not establish that the true covariance is zero (CCI Biomass, 2025b, pp. 14–22, 23–28). Change inference on fixed support requires the joint endpoint covariance structure and the difference of systematic endpoint errors. Uncertainty in forest classification, mixed-cell allocation and observation support remains separate.

Reporting comparisons require common cases and weights, together with coverage estimates for the full eligible population and pairwise overlap. Features derived from the target outcomes are distinguished from independently observed information, and algebraically redundant quantities are identified before interpreting an increment in reporting detail. Diagnostic comparisons require separate calibration and held-out evaluation, with uncertainty in estimated error rates. Practical-change thresholds and classification rules must be fixed without using held-out outcomes. Whole-pipeline calibration must account for spatial support, covariance, systematic bias and incomplete coverage, and comparisons with simple account, endpoint and trend baselines must use the same observations. No classification-performance estimates or physical change-resolution thresholds are inferred from the account availability criteria.

The public CSV, vector and XLSX reference representations linked by MapBiomas are examined separately for endpoint-label completeness, coordinate consistency, sampling-weight fields and explicit training indicators. A geographic reference location is not automatically an independently validated paired observation. Reference coverage is evaluated within the identical fixed Brazilian footprint. Blank reference entries are not interpreted as physical loss or zero biomass.

Stored native uncertainty is described separately from uncertainty calibration. On the fixed Brazilian rectangle, AGB and its SD are screened jointly against the delivered valid ranges, retaining valid zeros and distinguishing declared fill values. Counts and positive-AGB SD quantiles describe the stored values; they do not estimate error-correlation parameters or confidence

coverage. A separate reference-eligibility assessment considers released 500 m AGBref aggregates, target epochs, available census provenance, plot counts and published support-quality flags (Araza et al., 2026a,b). Shared coordinates or multiple target-epoch labels are not sufficient evidence of repeated field measurements.

3 Results

3.1 Availability of national account data

The workbook contains 13652 national records, 51 columns and 236 country or territory identifiers spanning 1961–2025. For 2015–2023, 183 identifiers satisfy the account availability criteria and 53 do not (Figure 1). All eight selected countries satisfy these criteria. The maximum discrepancy between the component sum and published total within the interval is 2 gha, within the rounding tolerance. These results describe account coverage rather than a physically matched analytical sample.

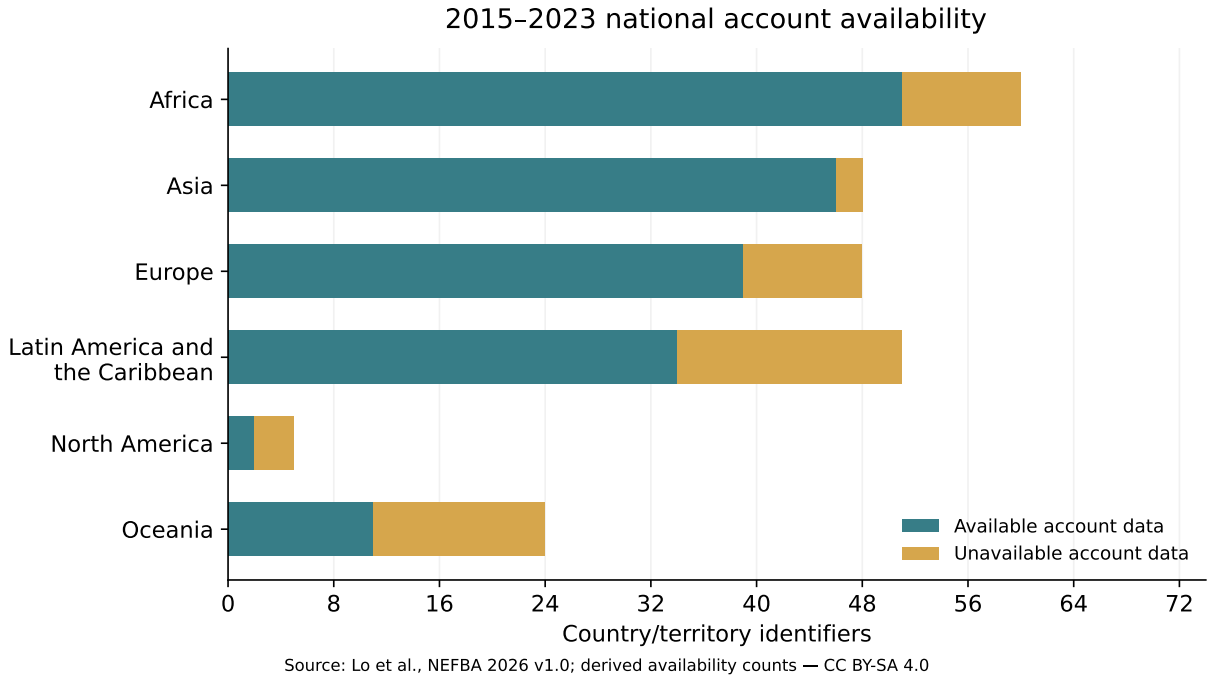


Figure 1: Availability of national biocapacity account data for 2015–2023 by source UN region. Availability requires complete annual coverage, appropriate publication flags, finite required values and numerical consistency. Data: Lo et al. (2026); derived availability counts distributed under CC BY-SA 4.0.

3.2 Territorial and endpoint compatibility

The candidate Canadian polygon extends to 83.14°N. Approximately 1.24% of its polygon area lies north of the biomass product's 80°N limit; the other seven candidate polygons lie within the documented latitude domain. These are geometric-domain results, not estimates of forest or biomass observation coverage. The biomass or forest content of the omitted Canadian support has not been established and cannot be assumed to be zero.

The French candidate polygon covers metropolitan-European coordinates, while the account workbook lists French Guiana, Guadeloupe, Martinique, Réunion and Mayotte separately. The Australian polygon includes outlying components. Both cases require component-specific territorial reconciliation rather than a sovereign-state label alone. The examined Copernicus satellite-

land-cover catalogue and retrieval form provide epochs through 2022, not the required 2023 endpoint (Copernicus Climate Change Service, 2026). This limitation applies to that product, not to every possible forest observation source. A 2022 classification cannot be substituted for 2023 without changing the comparison and assessing its consequences. No candidate has yet met all territorial, forest-support, complete-coverage and change-validation requirements for analytical admission.

3.3 Mapped cover on a common Brazilian support

The fixed footprint has area 61,830.34 ha and contains 65,536 native biomass cells. Each land-cover subset comprises 846 × 845 finer cells intersecting this rectangle. All target cells have complete published-label coverage at both endpoints under the declared unclassified-code convention. This is classification coverage within the local footprint, not national coverage or evidence that temporal gap filling was unnecessary.

Table 1 distinguishes the mapped quantities. The formations group occupies 6.99% in 2015 and 7.29% in 2023; including savannas gives 19.68% and 18.49%, respectively. The opposite endpoint directions concern different vegetation groupings and do not establish a resolved ecological change, a preferred forest definition or an account–asset divergence.

Table 1: Local mapped-cover endpoints on the identical Brazilian footprint. Values are area-overlap summaries of MapBiomas Collection 11 classifications, not independently validated forest-area estimates. The formations-plus-savanna row overlaps the first row and must not be added to it.

Mapped quantity	2015 (ha)	2023 (ha)
Forest formations	4,319.25	4,510.43
Formations plus savanna	12,169.71	11,435.33
Plantation (separate)	10.66	16.91
Open water (separate)	244.52	320.46

There are 13,301 native cells with a nonzero formations fraction at either endpoint and 10,385 at both endpoints, but only 625 are mapped entirely as formations at both endpoints. Mixed formations cells number 10,865 in 2015 and 11,154 in 2023. These descriptors show the difference between an area fraction and a whole-cell forest selection. Restricting the analysis to persistent pure cells would omit changing and mixed support; that restriction is not used to define a national forest-change sample.

3.4 Reference coverage and compatibility

The CSV and vector reference archives each contain 85,152 records but differ in temporal content and representation. The CSV has annual class fields through 2023, with only 4,053 nonblank entries for that endpoint. The vector table ends in 2022. The CSV uses textual class names, whereas the vector table uses numeric class representations; a compatible source-specific crosswalk is required before comparing those labels with Collection 11.

Seven reference locations occur within the fixed Brazilian footprint in each artifact, with coordinates verified against the vector point geometries. All seven CSV 2023 class entries are blank; the vector artifact has no such endpoint field. Neither examined artifact therefore supplies a local paired 2015/2023 reference sample. This is a limitation of these inputs, not evidence of absence of forest or ecological change. Collection 11's documented assessment and other possible reference sources are not invalidated by this finding.

The fully examined XLSX representation contains the same 85,152 identifiers as the CSV. At every identifier, all 117 common annual class, interpreter-count and border fields match after

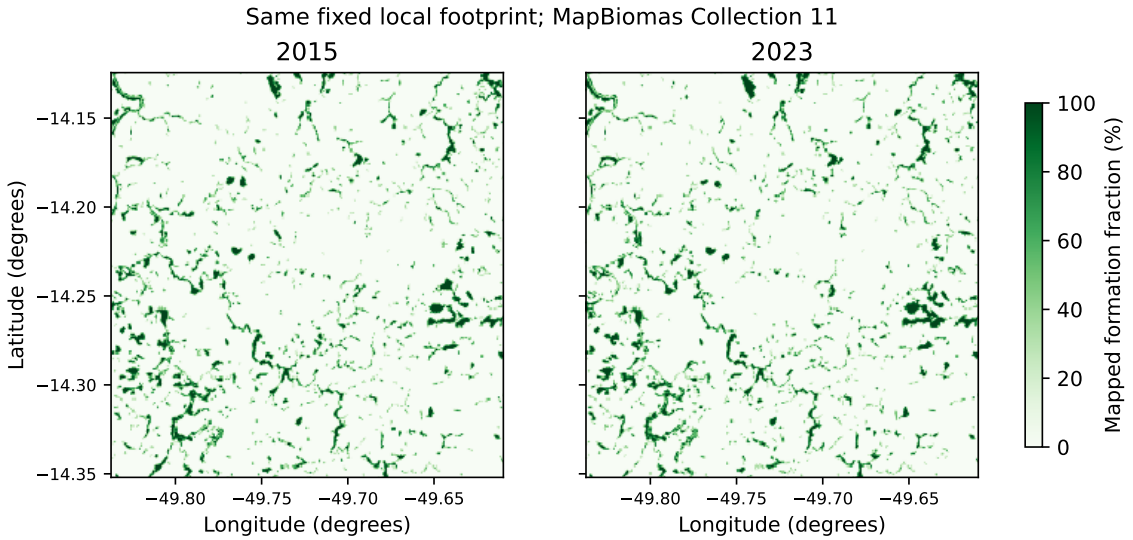


Figure 2: Mapped forest-formations fractions within the same fixed local rectangle in 2015 and 2023. Fractions refer to classes 3, 5, 6 and 49; savannas and plantations are separate. The complement includes other vegetation and land uses, not absence of ecological assets. The maps do not estimate biomass change or establish national representativeness. Source: MapBiomas Project, Collection 11; area-overlap processing described in the text.

whitespace, Unicode and integral-number serialization normalization. Coordinates and training-role tokens also match. The workbook therefore adds neither a new local paired-endpoint sample nor an independent reference replication.

The CSV explicitly marks 9,999 records as training samples. The remaining 75,153 are not thereby certified independent evaluation observations. Interpretation provenance, possible sample reuse, appropriate class/transition labels and probability-design treatment remain required. The Supplementary Information details coordinate-attribute and weight-field discrepancies that also preclude uncritical pooling of the artifacts.

3.5 Biomass support and uncertainty definitions

The inspected biomass aggregate contains 18 epochs on a 720×360 longitude–latitude grid with 0.5° spacing. The nominal 50 km designation does not imply equal-area cells. The Product User Guide defines aggregated density per total grid-cell area; it does not define a forest-area-normalized density. Its aggregate uncertainty calculation includes a spatial covariance component (CCI Biomass, 2026). Native biomass values likewise refer to full pixels and are not forest-masked. Consequently, the density denominator, valid-data support and uncertainty interpretation must be specified before a forest stock can be estimated. A finer mapped forest fraction does not identify the share of whole-pixel biomass inside that fraction.

The change product includes density difference, change standard deviation and quality-class layers. Quality classes are derived from the endpoint estimates and their uncertainty intervals, with additional growth-limit and spatial-filtering operations. The inspected example calculation of change standard deviation uses the sum of endpoint variances without a temporal covariance term. These classes therefore contain information about the processing rule and its inputs; they are not independent reference observations or calibrated 95% confidence statements. Supplementary Tables S2–S3 give the source definitions and associated interpretation requirements.

CCI's v7 stock validation includes the primary endpoint years and uses plot and LiDAR reference data reported as withheld from map production (CCI Biomass, 2026b, pp. 11–14, 64, 67). Reference preparation includes growth-rate adjustment for temporal mismatches of up to

ten years and forest-fraction harmonization of small plots (CCI Biomass, 2026b, p. 18). These stock comparisons provide relevant evidence about the observation system, but epoch-specific reference counts are not counts of paired change observations. The reported standard-error layer represents random error; the variance-consistency diagnostic does not estimate confidence-interval coverage for the local forest-stock difference (CCI Biomass, 2026b, p. 19).

Within the fixed Brazilian rectangle, 1,032 cells in 2015 and 946 in 2023 have positive AGB but stored SD equal to zero; their AGB estimates range from 1 to 4 Mg/ha. All 65,536 cells have valid AGB and SD at each endpoint. Median SD among positive-AGB cells is 6 and 7 Mg/ha, respectively. These are descriptive properties of integer-valued source layers, not evidence of zero true error. The source fields examined do not identify the origin of those zeros. Nor do the 44,021 cells with zero AGB at both endpoints establish persistent ecological absence (Supplementary Material, S7).

3.6 Biomass-reference eligibility

The open AGBref release contains one 500 m aggregate centroid inside the fixed Brazilian rectangle in each of its inspected 2010, 2015 and 2020 files (Araza et al., 2026b). Each local record has one contributing plot, a zero AGB estimate and zero minimum-plot, local-representativity and strict-filter flags. The descriptor requires more than one plot at 500 m; these records therefore fail that support criterion (Araza et al., 2026a). Their target-epoch labels do not disclose original census dates, and their different aggregate identifiers do not establish repeated original observations. The archive has no named 2023 member; an inspected unnamed file supplies no date field. This is accessible local reference information, but not a compatible paired 2015–2023 sample. No map–reference performance statistic is calculated from it (Supplementary Material, S9).

3.7 Conditional biomass-allocation sets

Table 2 reports the local allocation sets for the same two mapped groups. Both endpoint-difference sets contain zero, permitting negative, zero and positive differences under the minimal allocation model. Thus the sign is not identified by these point maps and nonnegativity alone, before considering measurement uncertainty. Adding savannas changes the mapped domain rather than supplying a confidence interval for the formations target.

Table 2: Conditional allocation sets in the fixed Brazilian footprint, in 10^3 Mg of above-ground woody biomass. Published cover and biomass point maps are treated as exact. Entries are compatible allocations under nonnegativity and unrestricted component density, not statistical confidence intervals or validated forest-stock changes. All difference bounds were calculated from unrounded endpoint bounds; displayed endpoints and differences are rounded independently.

Mapped domain	2015	2023	2023 minus 2015
Formations	$[44.23, 279.19]$	$[55.22, 326.23]$	$[-223.97, 282.00]$
Formations + savanna	$[229.75, 434.47]$	$[256.36, 489.29]$	$[-178.11, 259.53]$

The auxiliary sum of full-cell masses is 463.92×10^3 Mg in 2015 and 529.16×10^3 Mg in 2023. These totals refer to every native cell in the local rectangle, not only mapped forest; their increase does not identify a forest-specific gain and does not replace the primary forest-stock target. Observed zero-density cells remain included. The formations group has 791 and 876 mapped-pure cells at the two endpoints, with mixed cells retained in the upper sets. No within-forest mean-density point estimate or extent–density decomposition is inferred from these allocation sets.

3.8 Offsetting point-map differences on matched support

The auxiliary whole-window increase is 65.24×10^3 Mg. At native posting, positive and negative local mass differences sum to 142.21×10^3 and 76.97×10^3 Mg, respectively. Thus 70.23% of the sum of absolute local differences cancels algebraically in the net. These are differences between published cell-level point estimates, not measured gross growth and loss or an error-adjusted ecological-change estimate. Geographic cell areas, rather than nominal equal areas, are used throughout.

Coarsening preserves the net but reduces the observable offsets (Fig. 4a). At blocks of 64×64 native cells, every block difference is nonnegative and the cancellation fraction is zero, although the fine-grid negative differences remain in the underlying point maps. The result demonstrates information lost by this aggregation operation; it does not establish that every fine-grid difference is resolved by the sensor or distinguish ecological change from spatially correlated error.

3.9 Constructive sign ambiguity and information increments

The extremal witnesses attain the difference bounds in Table 2 while leaving all native-cell totals and mapped fractions unchanged. In particular, the same positive whole-window difference is compatible with a formations allocation difference of -223.97×10^3 Mg or 282.00×10^3 Mg. For the separately defined formations-plus-savanna domain the corresponding extremes are -178.11×10^3 and 259.53×10^3 Mg. Intermediate differences, including zero, are attainable by convex interpolation. Therefore even observing an aggregate biomass increase cannot, under this allocation model, rule out a mapped-group decline. This is a test of sign identification from the available point-map information, not evidence that either extreme occurred or that the two overlapping groups can attain their separate extrema jointly.

The coherent joint identified set is a hexagon rather than the rectangle formed from the separate marginal intervals (Fig. 3). Nevertheless, all four strict sign combinations remain feasible under a single three-component allocation at each endpoint. For example, one allocation gives formations -119.38×10^3 Mg and formations plus savanna 119.38×10^3 Mg; another gives 103.68×10^3 and -103.68×10^3 Mg, respectively. Thus the ambiguity is not an artifact of allowing incompatible biomass allocations to the nested groups. These are coherent hypothetical allocations, not evidence of observed group trajectories.

Table 3: Same-footprint information comparison under the unrestricted allocation model. Widths are in 10^3 Mg. Reduction compares native-cell mass observations with the one-window mass observation, retaining identical mapped domains and point maps. Neither information set identifies a group-change sign.

Mapped domain	One-window width	Native-cell width	Reduction (%)
Formations	993.09	505.97	49.05
Formations + savanna	993.09	437.64	55.93

The one-window information set has difference bounds $[-463.92, 529.16] \times 10^3$ Mg for either group, because both groups and their complements occupy positive area. Restoring native-cell mass observations narrows its width by 49.05% for formations and 55.93% for formations plus savanna (Table 3). Neither narrower set identifies a sign. The nine nested observation scales show that the width has already reached the one-window value at 32×32-cell blocks (Fig. 4b). This is an exact comparison of identification sets under matched support, not a held-out reporting-rule or classification-performance comparison.

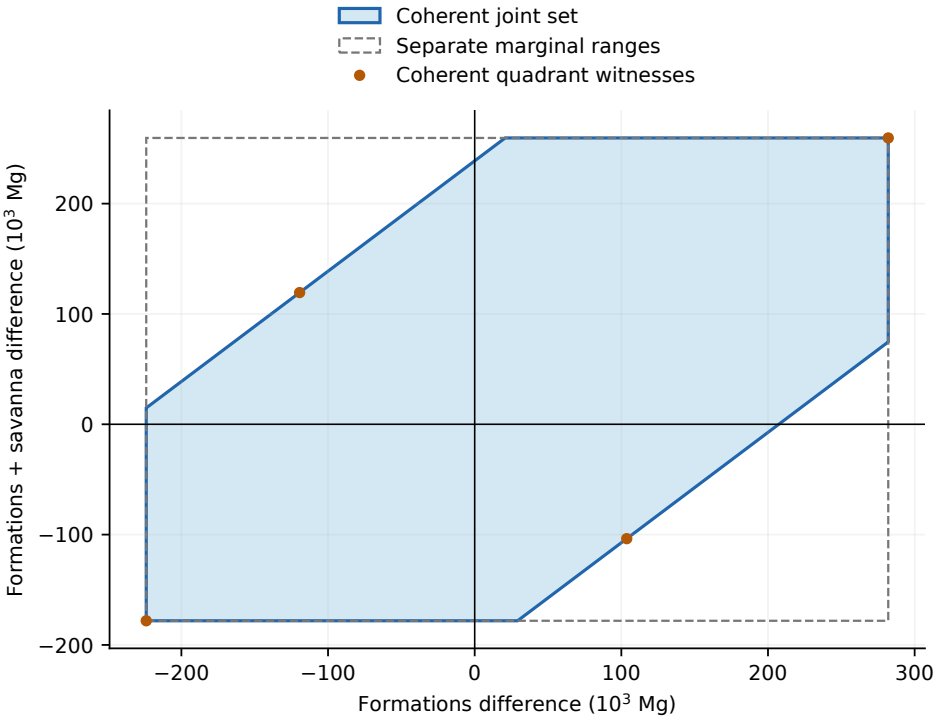


Figure 3: Sharp joint difference set for the nested mapped groups, conditional on exact point maps, nonnegative unrestricted allocation and no temporal coupling. The added-savanna constraint cuts off combinations admitted by the rectangle of marginal ranges. Four reconstructed allocations attain every strict sign quadrant while preserving all native-cell totals and component-presence geometry. Points are feasible allocation witnesses, not ecological observations or confidence regions.

3.10 Allocation restrictions required for a positive sign

Within the density-contrast sensitivity family, homogeneous allocation ($R = 1$) gives positive model differences of 28.78×10^3 Mg for formations and 41.87×10^3 Mg for formations plus savanna. These are assumption-dependent outputs, not empirically identified forest-stock estimates. The lower difference bound reaches zero at $R = 1.4555$ and $R = 1.9495$, respectively (Fig. 4c). Strict positive-sign identification within this family therefore requires a common contrast bound below the corresponding value; at the boundary zero is permitted. Neither contrast bound has been measured or validated here. The unrestricted sets remain the primary results. The sensitivity analysis identifies a specific additional information requirement rather than supplying it by assumption.

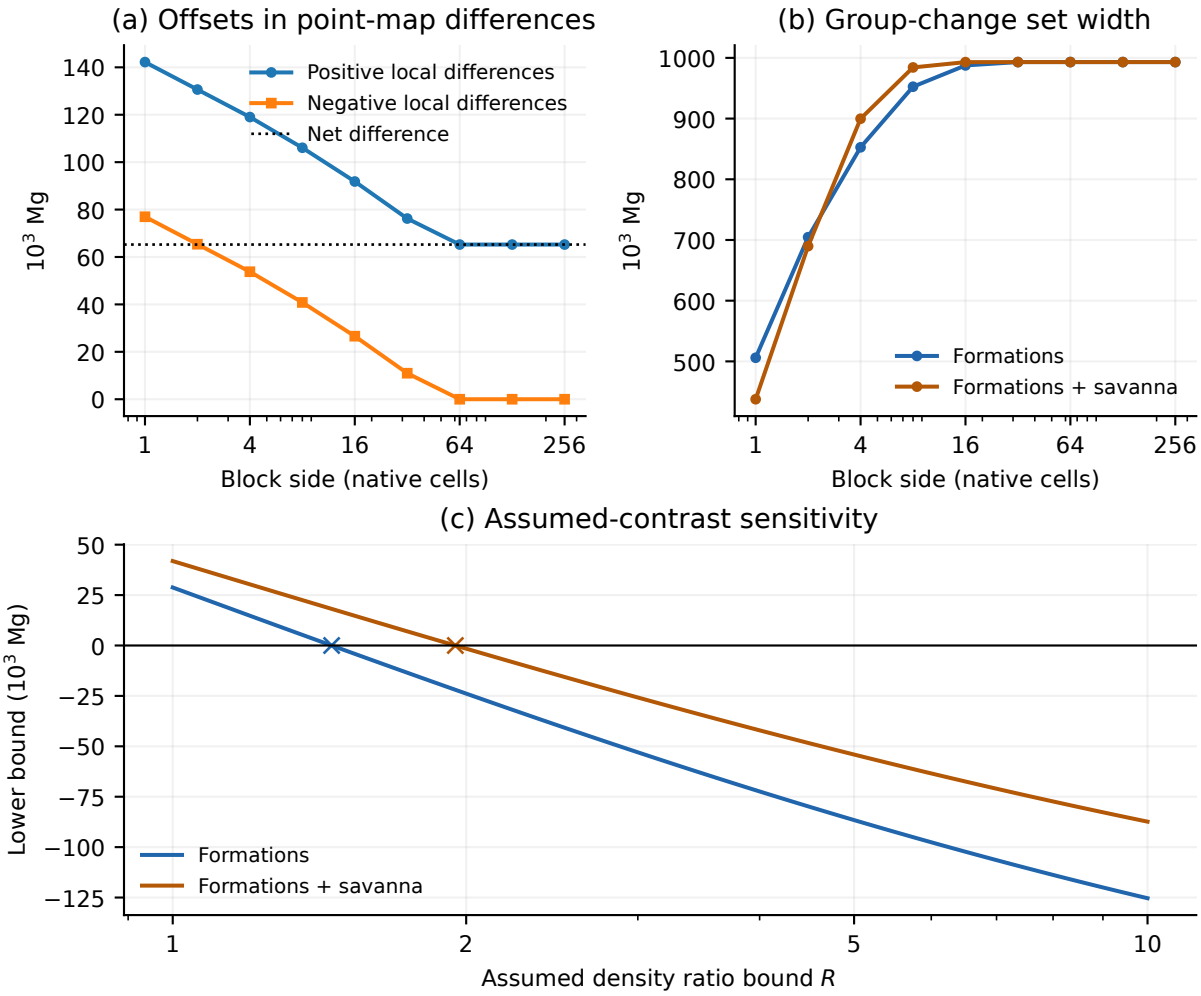


Figure 4: Fixed-support point-map information analysis. (a) Summed positive and negative local mass differences decrease under spatial aggregation while the net is preserved. (b) Mapped-group difference-set widths increase as native-cell mass observations are replaced by block totals; cover definitions and footprint remain fixed. (c) The lower difference bound under an assumed within-cell density-contrast limit R; crosses mark zero-boundary values. No finite R is empirically established. Native posting is not independent effective resolution. Panels describe conditional point-map arithmetic and allocation information, not calibrated ecological change, national trends or classification accuracy.

4 Discussion

The account coverage supports a multi-country comparison, but availability alone does not establish matched physical support. The distinction is consequential where country definitions, forest masks, mapped observation domains or density denominators differ. Whole-cell biomass averages and forest-only means require different area multipliers. Applying the same multiplier to both would change the physical quantity being estimated.

Spatial compensation also requires measurement-specific interpretation. Opposing local net changes can occur within an increasing national total, but spatial error can produce the same pattern in point estimates. Cancellation must therefore be evaluated with spatial covariance, resolution and zero-change error models. A diagnostic evaluated against a target-derived quality class would measure agreement with that processing rule, rather than independently verified ecological change.

The evidence distinguishes arithmetic cancellation in delivered point maps, mapped-group non-identification, and ecological composition masking. The first is quantified on common support, and the second is established by constructive allocations with opposite signs. The third requires independently supported account–asset outcomes and is not demonstrated by either result. The matched-information comparison measures identification-set narrowing, not reporting-rule accuracy. National forest-stock trends, account–asset divergence and conversion associations are not estimated. National inference requires territorial and mask reconciliation, independently justified change uncertainty, and coverage-sensitive treatment of missing or converted pixels. Soil, wetlands and fisheries involve different observation domains and are outside the present assessment.

The Brazilian example separates mapped forest formations, savannas and plantations rather than treating all woody vegetation as one invariant target. Temporal filters and gap filling can affect endpoint timing, while map-wide classification accuracy does not determine local or transition accuracy. The public reference-point downloads inspected are labelled Collection 9; compatibility with Collection 11, sample reuse, design weights and transition-specific reference interpretation require assessment before change validation. Different map publishers do not establish independent errors, particularly where auxiliary information or reference data may overlap.

The biomass observation operator imposes a further distinction. Even an exactly known forest fraction does not allocate total woody biomass between forest and other parts of a mixed pixel. A whole-territory woody-biomass stock is a different estimand from forest-only stock; neither should be relabelled as the other. Extent–density decomposition requires the stock and forest-area operands to describe the same support. Multiplying native density by mapped forest fraction identifies no forest-only biomass estimate without an allocation assumption; the homogeneous sensitivity endpoint supplies such an assumption, not independent evidence. The computed allocation sets quantify one part of the identification problem while conditioning on uncertain point maps; they do not supply a calibrated change-resolution rule. The contrast boundaries make an additional information requirement explicit: within this family, positive-sign identification requires a uniformly small enough density contrast in all mixed cells and at both endpoints. Independent measurements or defensible ecological constraints could test such a restriction, but neither increased map detail nor an assumed contrast substitutes for paired change validation.

The provider validation and the present identification limits address different questions. A held-out stock comparison can assess map agreement on its reference support without identifying forest-specific biomass inside a mixed cell or calibrating a paired 2015–2023 change on the present footprint. The described validation screening excludes plots deforested between inventory and map dates, and temporal harmonization uses growth assumptions; transferring its error characteristics to converted land requires evidence beyond agreement among retained forest observations (CCI Biomass, 2026b, pp. 13, 18–19). Reference measurement error, geolo-

cation and within-pixel sampling also contribute to map–reference discrepancies. Accordingly, neither RMSD nor a globally pooled bias pattern is adopted as a local change-error model, and the absence of such a model here is not a claim that the v7 stock product lacks validation.

5 Conclusions

National biocapacity data for 2015–2023 satisfy the specified availability criteria for 183 of 236 identifiers, while all eight screened national physical comparisons remain on hold. In the matched Brazilian footprint, an increasing whole-cell biomass point-map total coexists with offsetting local differences and with both signs of mapped-group change under the unrestricted allocation model. Native-cell mass information narrows identification sets substantially but does not identify a sign. Conditional density-contrast boundaries identify a possible auxiliary-information route to sign identification without asserting that the requisite bounds hold. These results establish limits and increments of identification from the available observations, not measured national composition masking. Boundary matching, independently justified change uncertainty and paired reference evidence remain necessary for the broader ecological account–asset test.

Data and code availability

Source datasets are cited in the text. Derived account-availability tables, source definitions, file hashes and computational methods are available at https://github.com/MIKEAA2020/general-sustainability/tree/main/batch%209/agent%202/ecomod_empirical_followup/F11_A01. NEFBA-derived tables and figures are distributed under CC BY-SA 4.0. Boundary inputs and boundary-derived data retain their separately documented source licences and attribution requirements. Conditional allocation summaries, topological intersection counts, matched-scale comparisons, aggregate witness certificates and density-contrast sensitivity results are also supplied; raw native-density arrays are not redistributed. MapBiomas-derived cover summaries identify Collection 11 and the processing used; the provider permits public use with source attribution. Raw country land-cover images, biomass files and provider technical reports are not redistributed.

Declaration of generative AI in scientific writing

Arena.ai Agent Mode was used to assist with computational methods and literature review.

References

Abaee, A. (2026). A rising biocapacity total can mask ecological-capital land drawdown: a two-land model. Manuscript. https://github.com/MIKEAA2020/general-sustainability/tree/main/batch%209/agent%202/ecomod_c17.

Mancini, M. S., Galli, A., Niccolucci, V., Lin, D., Hanscom, L., Wackernagel, M., Bastianoni, S., and Marchettini, N. (2017). Stocks and flows of natural capital: Implications for Ecological Footprint. *Ecological Indicators*, 77, 123–128. https://doi.org/10.1016/j.ecolind.2017.01.033.

Maes, J., Bruzón, A. G., Barredo, J. I., Vallecillo, S., Vogt, P., Rivero, I. M., and Santos-Martín, F. (2023). Accounting for forest condition in Europe based on an international statistical standard. *Nature Communications*, 14, 3723. https://doi.org/10.1038/s41467-023-39434-0.

Olofsson, P., Foody, G. M., Herold, M., Stehman, S. V., Woodcock, C. E., and Wulder, M. A. (2014). Good practices for estimating area and assessing accuracy of land change. *Remote Sensing of Environment*, 148, 42–57. https://doi.org/10.1016/j.rse.2014.02.015.

Lo, K., Basnet, N., Miller, E., Srikanthalingam, B., Foley, B., Van Berkum, J. L., Long, A. H., Toneva, P., Ermina, M., and Klinkenberg, C. (2026). *National Ecological Footprint and Biocapacity Accounts, Edition 2026 (1961–2025): Results and metadata, version 1.0*. Footprint Data Foundation. https://footprint.info.yorku.ca/data/.

Santoro, M., and Cartus, O. (2026). ESA Biomass Climate Change Initiative: Global datasets of forest above-ground biomass for 2005–2012 and 2015–2024, v7.0. CEDA. https://doi.org/10.5285/6429d1aafe1e43b9b414e4a5a7f8b903.

CCI Biomass (2026). *Product User Guide v7.0*. 19 March 2026. https://climate.esa.int/en/projects/biomass/key-documents/.

Runfola, D., et al. (2020). geoBoundaries: A global database of political administrative boundaries. *PLOS ONE*, 15, e0231866. https://doi.org/10.1371/journal.pone.0231866. Country-specific geometry sources, represented years and commits: https://www.geoboundaries.org/api.html.

Copernicus Climate Change Service (2026). *Land cover classification gridded maps from 1992 to present derived from satellite observations*. Catalogue and retrieval form accessed 29 October 2026. https://doi.org/10.24381/cds.006f2c9a.

MapBiomas Project (2026). *Collection 11 of the Annual Land Cover and Land Use Maps of Brazil*. General Algorithm Theoretical Basis Document, class codes and descriptions, and biome appendices. Accessed 29 October 2026. https://brasil.mapbiomas.org/downloads/.

CCI Biomass (2025a). *Algorithm Theoretical Basis Document, v7.0*. 17 December 2025. https://climate.esa.int/media/documents/Algorithm_Theoretical_Basis_Document_ATBD_V7.0.pdf.

CCI Biomass (2025b). *End to End ECV Uncertainty Budget, v7.0*. 17 December 2025. https://climate.esa.int/documents/3354/End-to-End_ECV_Uncertainty_Budget_E3B_V7.0.pdf.

CCI Biomass (2026b). *Product Validation and Intercomparison Report, v7.0*. 24 April 2026. https://climate.esa.int/documents/3356/Product_Validation_Intercomparison_Report_PVIR_V7.0.pdf.

Araza, A., Herold, M., Avitabile, V., et al. (2026a). AGBref: A global reference dataset of above-ground forest biomass. *Research Square*, preprint, posted 9 January 2026. https://doi.org/10.21203/rs.3.rs-8211898/v1.

Araza, A., Bielig, C., and Flores, R. A. A. (2026b). AGBref—Global Reference Dataset for Above-Ground Biomass (AGB). Zenodo, record 20793072, file `agb-ref-v2.zip`. https://doi.org/10.5281/zenodo.20793072.